



The HVAC Revolution

Carrier's brand-new Infinity line answers homeowner demand for a resilient HVAC system that offers the highest quality indoor air, comfort, and unparalleled energy efficiency.



Turn to the experts



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BEING A LEADER IN ANY INDUSTRY requires a revolutionary mindset—but being revolutionary doesn't always mean rolling out a brand-new product or even inventing a product that didn't formerly exist. Sometimes it is all about strategic enhancement of existing workhorse products and precision knowledge about what consumers want.

HVAC leader Carrier recently added technological upgrades and enhanced workmanship to its popular Infinity HVAC product line. As this article will detail, the company completely transformed the product line into a truly superior solution for the residential home market.

In fact, ever since its launch in 2011, the Infinity heat pump and air conditioner with Greenspeed Intelligence system has been the superior HVAC choice. “These products pushed performance and heating efficiency for an air-source heat pump to the forefront of the industry,” says Lynn Turner, Carrier’s associate director for residential cooling platform and product marketing.

Today, this intelligent inverter-controlled HVAC system has been redesigned and upgraded with features not available on any competitor system, notes Turner. The new Infinity 26 Air Conditioners and Infinity 24 Heat Pumps with Greenspeed Intelligence are indeed the gold standard for residential heating and cooling.

Let’s walk through the four main components that make Carrier’s Infinity with Greenspeed Intelligence the perfect solution for today’s high-performance homes.

Gold Standard HVAC System

Here, the HVAC specialists at Carrier spec the ultimate HVAC system.

"The Infinity is already the top of the line for air conditioning and heat pump technology—the pinnacle of HVAC," says Todd Nolte, senior director, Product Strategy and Regulatory, HVAC-Residential, Carrier. "To this you could add customization with the best add-ons for air purification, zoning control, an ERV/HRV, and a humidifier. This would give you the highest efficiency and the highest level of comfort and control."

Here's what a premier system would include:



The Infinity 26 Air Conditioner with Greenspeed Intelligence, the Infinity 98 gas furnace, and the Infinity System Control.

The ERVXXLHB

Infinity Air Purifier

**Ultimate
HVAC
System**

ENERGY EFFICIENCY BEST IN CLASS

HVAC manufacturers continually innovate, test, and redesign to add ever-higher efficiency to components that are already very efficient. Carrier's high-efficiency products stand out because of their use of variable-speed technology.

Variable speed is not well-known with consumers, who have been trained to look at SEER numbers as the ultimate comparative metric when buying HVAC systems. The car analogy helps. "How many people have a car with one gear?" Turner asks. No one, of course, because they don't exist. HVAC in similar fashion runs smarter when it is matching its output to the changing demands of the house.

With the new Infinity line, the variable-speed technology changes speed hourly, daily, weekly, and yearly to match the conditions of the home and the occupants' habits. A single-stage unit simply can't offer the same level of customized comfort, Turner notes.

Let's start with energy efficiency. "These products offer 13 EER or better across the lineup, including the industry's only 13 EER 5-ton variable-speed ducted heat pump and the highest 3-, 4-, and 5-ton SEER and 5-ton EER variable-speed air conditioner efficiencies," Turner says.

The line also offers up to a 26 SEER/16.5 EER 2-ton variable-speed air conditioner with specific size and equipment combinations.

"Carrier's residential heating and cooling equipment has some of the highest cooling efficiency ratings in the AHRI directory for SEER and EER," Turner says. "We also have the highest ducted variable speed heating efficiency ratings up to 13 HSPF."

These efficiencies not only translate into cost savings for homeowners, but they also mean a higher level of comfort in the home.

LEADERSHIP IN SOUND REDUCTION

While the Infinity product line is running in the background, it does not add to the ambient noise of the house. "We have the lowest sound level in the ducted industry—51 decibels, which is quieter



“Our new 4-ton Infinity units can match the home’s load requirements down to 25% capacity—the equivalent of a 1-ton unit. That’s kind of like having four units in one and it’s a significant improvement of the 40% of capacity operation offered by our previous models.”

—Lynn Turner, associate director for residential cooling platform and product marketing, Carrier

than a normal conversation,” Turner says.

This concept can be hard for consumers to grasp. While they understand quieter is better, the metrics for representing sound aren’t as clear. Educating homeowners on how the scale for logarithmic sound is essential.

“The next closest sound level available to consumers is 54 decibels, and we are at 51 decibels,” Turner says. Three decibels doesn’t sound like much, but it makes a huge difference. Those three decibels cut the noise in half, bringing the sound in line with the level of noise created by normal indoor conversation.

The reason the new Infinity line is whisper quiet is because of the quality components that make up the system. “The combination of the different parts is what sets us apart,” Turner explains. “This includes a smooth-running compressor, the Silencer System II, which includes the Silencer top cover and discharge grille, integrated fan motor, forward-swept fan blades, compressor vibration isolator plate, two compressor sound hoods and

split-post compressor grommets that maximize airflow, minimize sound, and minimize vibration.”

Another option to keep noise levels low in a home is the Quiet Mode on the Infinity 24 heat pumps. The function allows homeowners to cap the sound at 69 decibels to ensure a quiet outdoor unit—and a peaceful night’s sleep.

NIMBLE TECHNOLOGY RESPONSE

Carrier’s new Infinity line offers a technological sophistication that benefits homeowners, builders, and technicians. The Bluetooth connectivity built into the system and the Wi-Fi-connected Infinity System Controls mean the units are easier to service, and software upgrades can happen remotely.

“Our ability to perform advanced diagnostics and interactive troubleshooting is having a positive impact for us with the technician shortage,” Turner says. Formerly, technicians had to go out to a home, assess the problem, and then return with parts to make the repair. With remote assessment, technicians can be

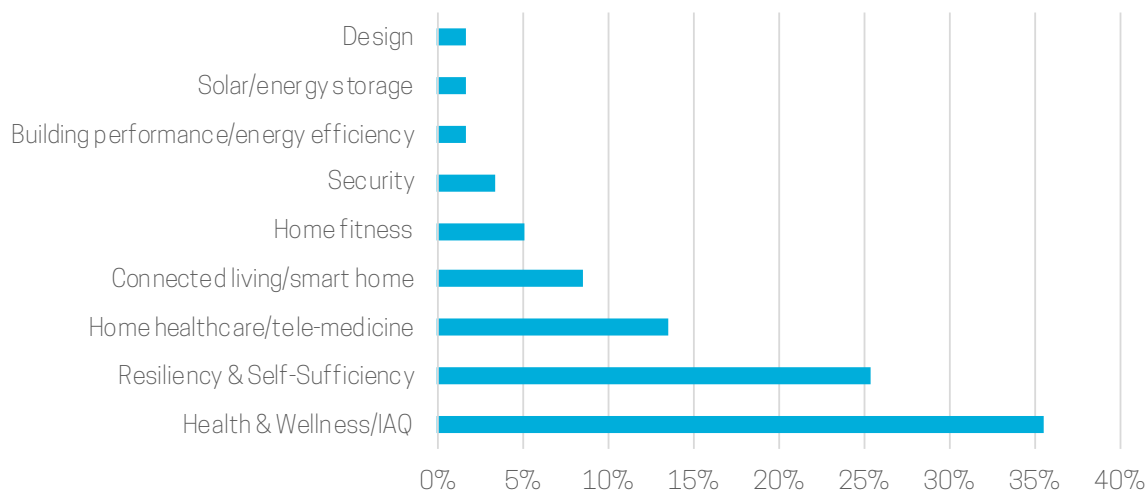
more efficient and focus on what they do best—keeping Carrier products updated and running smoothly.

The upside to the homeowner is obvious; no one wants to stay home to let in a technician multiple times, and the flexibility of this technology presents the benefit of contactless service, as well. There are also advantages for the builder, who can point out the convenience factor to prospective buyers when illustrating the ongoing benefits of owning a Carrier HVAC system.

The Infinity models include more than 130 specific-source points of diagnostics, which can be accessed through a new app or through the wall control in the home. The app also includes “fault trees” for interactive troubleshooting assistance, again, facilitating quick issue resolutions.

“Never before have we been able to upgrade the software of the units,” says Turner. “In the past, if there was an upgrade we had to change the unit’s board. That’s like having a smartphone and when new updates come along having to throw the phone away,” she explains.

IAQ: Top of Mind



During the first quarter of 2020, COGNITION Smart Data tracked consumer interest in home-related categories. [Health & Wellness/IAQ was the most talked about topic.](#)

Infinity Air Purifier: Clean Air When It Counts

Today's home buyers are focused on healthy indoor air quality.

Getting pure, safe air starts with choosing the best air purifier.

Research shows that consumers are increasingly interested in indoor air quality (IAQ) and are demanding high-quality air in their homes and other indoor environments.

"Companies in the home building space are moving forward on enhanced innovations for their products to help homeowners adapt to today's living realities," says Green Builder Media CEO Sara Gutterman, author of a 2020 report on IAQ. "There are new requirements everywhere, from making the air in homes healthier, strategies to make homes virus resistant, and at the same time bringing in technological innovations to make it seamless and easy to use for the homeowner."

According to this research, indoor air quality, and health and wellness are far outpacing interest in other home factors, such as design. (See chart, IAQ: Top of Mind.) This is not surprising, as consumers had already demonstrated a strong interest in indoor-outdoor spaces and wellness trends over the past five years.

According to the EPA, the level of pollutants in the air inside homes is two to five times

"Building a system featuring Carrier top-tier products as a complete Infinity System provides homeowners with a new level of home comfort," Nolte says. "Plus, our advanced system control allows the homeowner to synchronize their comfort needs and receive energy reports to better maximize savings. And the Infinity air purifier would provide the perfect complement to our Infinity heating and cooling equipment by providing cleaner, fresher air in the home."

[Learn more and find a dealer near you!](#)



higher than outdoors. These pollutants are a function of our lifestyles, the products we use, and the materials used to build our homes. With respiratory issues on the rise, heightened awareness of viruses, and increased awareness of the effects of overall health, more homeowners are interested in solutions to treat their air.

Carrier has the ultimate solution with its Infinity Air Purifier at MERV 15 and with a patented Captures & Kills technology, which provides a high tested germicidal effectiveness with a 99% activation rate against select viruses and bacteria.

The system uses a three-step process to create clean air:

1. Precision point ionization charges particles as they enter the purifier.
2. A specially designed, continuously charged

media filter captures airborne particles.

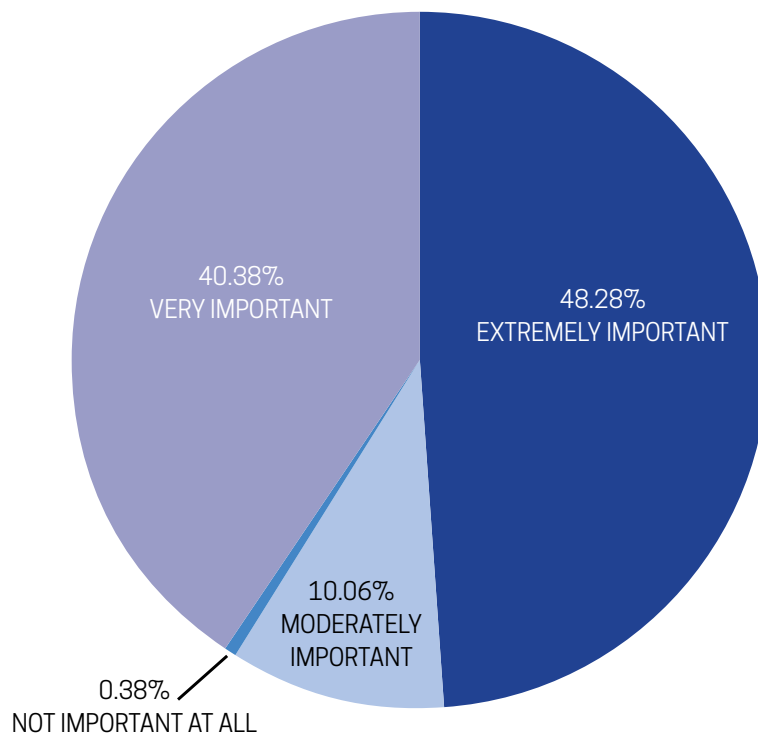
3. Patented, state-of-the-art technology kills select captured viruses and bacteria* and can prevent further growth of bacteria and fungi on the filter.

"Our air purifier is our most effective filtration to treat the air in your home," says Turner. "The filter can catch bacteria and select virus particles. Then the electrical field kills it to ensure there is no chance it is distributed in the home. Typically, products that have this type of performance cause increased stress on the components in your furnace. Our air purifier treats the air while protecting the reliability of the equipment it is attached to."

[Learn more about the Carrier Infinity Air Purifier.](#)

*Based on third-party testing (2012) showing 95% of particles size 1.0 to 3.0 microns captured. Based on third-party testing (2012, 2007) showing >99% kill rate for common cold surrogate, *Streptococcus pyogenes*, and human influenza.

How Important Is IAQ to You?



Before the coronavirus outbreak (fourth quarter 2019), Green Builder Media surveyed [consumers on IAQ topics](#). Nearly nine of ten (88%) respondents indicated IAQ was “extremely important” or “very important” to them.

HUMIDITY CONTROL POWERHOUSE

The new Infinity products were designed with dehumidification squarely in mind, but the truth is, many people don’t really understand how humidity and the conditioning of air are tied together.

Higher humidity can affect homeowner comfort by reducing the body’s natural ability to cool itself, so removing additional indoor humidity can make a home feel cooler and potentially allow occupants to increase the thermostat setting in the warmer months. A properly dehumidified home will feel more comfortable, and the utility bills will be lower.

“Greenspeed Intelligence with inverter-controlled, variable-speed operation is like having cruise control for comfort,” says Turner. “It offers a steady, consistent temperature and can remove up to 400



Wi-fi connected Infinity System Control

percent more moisture than standard systems.”

The units do this in a variety of ways, including:

- A wider operating range down to 25% of capacity, which can reduce cooling

output to dehumidify more without overcooling.

- Longer run times for more time removing moisture.
- Adjusted compressor and blower speeds to achieve targeted capacity and humidity levels.
- Less cycling during cool and damp conditions, which means longer run times and more moisture removal when a single-stage unit would cycle on and off.

According to Carrier testing, a Greenspeed unit removes up to 70% more moisture than a single-stage 13 SEER system under Association of Home Appliance Manufacturers (AHAM) conditions and up to 30% more moisture than Carrier’s current Greenspeed units under the same conditions.

THE NEXT GENERATION OF BENEFITS

The new Infinity air conditioners and heat pumps with Greenspeed Intelligence have a lot to offer builders, installers, and homeowners.

This chart summarizes the main feature benefits of the new Infinity air conditioners and heat pumps with Greenspeed Intelligence. If you are a builder, use the homeowner benefits in your marketing collateral to illustrate how your Carrier-outfitted homes offer better air, energy efficiency, reliability, and comfort.

Infinity® air conditioners and heat pumps with Greenspeed® intelligence have a lot to offer for dealers and homeowners alike. To help summarize, we've taken the highlights and put them into a chart you can refer to before talking with your customers.

FEATURE	HOMEOWNER BENEFIT
Inverter-controlled variable-speed compressor with Greenspeed intelligence	■ Like having cruise control for comfort – more precise temperature and humidity control ■ Steady, consistent comfort with fewer temperature swings ■ Expected longer unit life due to less wear and tear on equipment ■ Improved air quality within the home due to longer run times
Market-Leading Efficiency on Most Models ¹	■ Long term savings on cooling/heating costs ■ Potential utility company rebates
Market-Leading Low Sound ²	Enjoy a backyard barbecue or conversing outside with neighbors with quiet, unobtrusive comfort
Quiet Mode (via the Infinity® System Control)	Enjoy a peaceful night's sleep with quiet, unobtrusive comfort even if the unit is under the bedroom window
Outstanding Dehumidification	Due to longer run times, the homeowner will feel more comfortable with lower humidity in the cooling season, and can save on their energy bill with a higher thermostat set point
Operation Down to 25% of Capacity	■ Only pay for the cooling/heating you need with comfort more precisely matched to conditions ■ Quieter operation ■ Less wear and tear on equipment
Expanded High and Low-Ambient Performance	Enjoy continued comfort and efficiency, even during extreme temperatures
Long Line Set Length	Unit installation where it works for you—away from bedroom windows or backyard patios
5-Ton Unit Advantage ¹	Enjoy energy saving efficiency and performance normally reserved for smaller capacity units
Enhanced Service App	Streamlined service call allows homeowner to get back to their regularly scheduled day
More Direct-Source Diagnostic Codes	Zero in on service needed
Detailed Troubleshooting Trees	Streamlined service call allows homeowner to get back to their regularly scheduled day
Bluetooth Pairing with Outdoor Unit	Potentially less intrusion with fewer tech visits inside the home
Over-the-Air Software Updates	Enjoy incredible comfort today, and take advantage of available performance improvements down the road

¹ Depending on model combination. Based on AHRI EER efficiency ratings compared to competitive 5-ton variable-speed heat pumps, and SEER/EER on 3-, 4- and 5-ton variable-speed air conditioners, as of February 2020.

² Depending on model combination. Based on laboratory testing of the Infinity® 26 air conditioner and Infinity® 24 heat pump models by Carrier engineers in Fall 2019 and AHRI sound level data for competitors' variable-speed outdoor units, published as of December 2019.



“Building a system featuring our top-tier products as a complete Infinity System, coupled with an Infinity air purifier, would certainly provide homeowners with a new level of home comfort.”

—**Todd Nolte, senior director,
Product Strategy and Regulatory,
HVAC-Residential, Carrier**

ATTENTION TO DETAIL

Carrier is proud of the four areas of excellence embodied in the revamped Infinity line, but there are other upsides to the products as well.

“Our new 4-ton Infinity units can match the home’s load requirements down to 25% capacity—the equivalent of a 1-ton unit,” Turner says. “That’s like having four units in one, and it’s a significant improvement of the 40% of capacity operation offered by our previous models.”

This upgrade means that the system is quieter because it is operating at reduced compressor and fan speeds, there will be less wear and tear on the unit with fewer cycles, the air quality will be improved with more filtering time, and higher efficiency will result during lower load requirements.

Another asset is the enhanced operating range, which makes the units suitable for extreme climate. The new Infinity products maintain about 80% capacity at temperatures higher than 125 degrees F, which is similar to single-stage units in desert conditions. On the low-ambient heating range, the air-source ducted heat

pump system delivers comfort down to minus 15 degrees F.

LOOKING TO THE FUTURE

The new Infinity 26 air conditioners and Infinity 24 heat pumps with Greenspeed Intelligence meet the highest demands of today’s homes. However, when Carrier looked at how to make the Infinity better, it didn’t just innovate on the technology and systems front, it also considered how the products are sold by builders, serviced by technicians, and ultimately enjoyed by homeowners across the country and in all climates.

Turner notes that the rollout of this HVAC line this spring signaled a new era of intuitive understanding about how today’s homeowners want to live in their homes. “This really is a huge change in technology, and it revolves around the intelligence of how the systems communicate. It’s the next generation of inverter-controller comfort, and it will come to the consumer at the same price point as a typical top-of-the-line option—but with so much more to offer.” ■

[Learn more about the Carrier Infinity with Greenspeed Intelligence.](#)

Why Does Energy Efficiency Matter?

These facts prove energy efficiency isn’t just about lowering utility bills

- In 2018, energy-efficiency investments totaled \$11 Billion in the United States.
- Many markets across the United States offer [funding, rebates, and incentives for energy efficiency](#).
- Enhanced mandates and codes on the state, city, and municipality level demand energy-efficient building. [This webinar gives an overview of code changes.](#)
- Zero energy organizations and voluntary energy-efficiency programs are on the rise (Energy Star, Passive House, Zero Net Energy Ready, EEBA, HERS, NBI and others)
- Utilities, municipalities, and cities are implementing demand-side energy management programs, particularly in California, New York, Vermont, Washington, Oregon, Colorado, Minnesota, and Utah.
- Climate change-caused weather extremes are impacting indoor comfort, which is a major factor in consumers investing in high-performance products like HVAC.

Source: “Essential Trends to Grow Your Business” 2020, [COGNITION Smart Data](#)